

ECEN 629 Applied Convex Optimization

Chapter 1. Introduction

Instructor: Dr. Chao TIAN

Texas A&M University-College Station

Optimization vs. Mathematical Optimization

Optimization

- Intuitively: find the best way for something.

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- Intuitively: find the best way for something.

Mathematical optimization

- A rigorous formulation of the practical problem;
- Represent mathematically as a set of variables, an objective function, and a set of constraints.

Mathematical optimization

(mathematical) optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \dots, m\end{array}$$

- $x = (x_1, \dots, x_n)$: optimization variables
- $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$: objective function
- $f_i : \mathbf{R}^n \rightarrow \mathbf{R}, i = 1, \dots, m$: constraint functions

optimal solution x^* has smallest value of f_0 among all vectors that satisfy the constraints

Examples

portfolio optimization

- variables: amounts invested in different assets
- constraints: budget, max./min. investment per asset, minimum return
- objective: overall risk or return variance

device sizing in electronic circuits

- variables: device widths and lengths
- constraints: manufacturing limits, timing requirements, maximum area
- objective: power consumption

data fitting

- variables: model parameters
- constraints: prior information, parameter limits
- objective: measure of misfit or prediction error

An Exercise

“Optimization”

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Mathematical optimization

- What are the variables?
- What is the objective function?
- What are the constraints?

Solving optimization problems

general optimization problem

- very difficult to solve
- methods involve some compromise, *e.g.*, very long computation time, or not always finding the solution

exceptions: certain problem classes can be solved efficiently and reliably

- least-squares problems
- linear programming problems
- convex optimization problems

Least-squares

$$\text{minimize } \|Ax - b\|_2^2$$

solving least-squares problems

- analytical solution: $x^* = (A^T A)^{-1} A^T b$
- reliable and efficient algorithms and software
- computation time proportional to $n^2 k$ ($A \in \mathbf{R}^{k \times n}$); less if structured
- a mature technology

using least-squares

- least-squares problems are easy to recognize
- a few standard techniques increase flexibility (*e.g.*, including weights, adding regularization terms)

Linear programming

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

solving linear programs

- no analytical formula for solution
- reliable and efficient algorithms and software
- computation time proportional to n^2m if $m \geq n$; less with structure
- a mature technology

using linear programming

- not as easy to recognize as least-squares problems
- a few standard tricks used to convert problems into linear programs
(*e.g.*, problems involving ℓ_1 - or ℓ_∞ -norms, piecewise-linear functions)

Convex optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \dots, m\end{array}$$

- objective and constraint functions are convex:

$$f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$$

if $\alpha + \beta = 1$, $\alpha \geq 0$, $\beta \geq 0$

- includes least-squares problems and linear programs as special cases

solving convex optimization problems

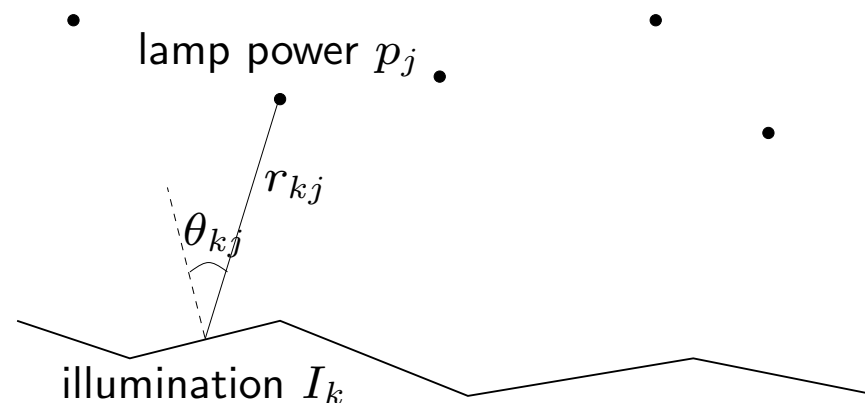
- no analytical solution
- reliable and efficient algorithms
- computation time (roughly) proportional to $\max\{n^3, n^2m, F\}$, where F is cost of evaluating f_i 's and their first and second derivatives
- almost a technology

using convex optimization

- often difficult to recognize
- many tricks for transforming problems into convex form
- surprisingly many problems can be solved via convex optimization

Example

m lamps illuminating n (small, flat) patches



intensity I_k at patch k depends linearly on lamp powers p_j :

$$I_k = \sum_{j=1}^m a_{kj} p_j, \quad a_{kj} = r_{kj}^{-2} \max\{\cos \theta_{kj}, 0\}$$

problem: achieve desired illumination I_{des} with bounded lamp powers

$$\begin{array}{ll} \text{minimize} & \max_{k=1, \dots, n} |\log I_k - \log I_{\text{des}}| \\ \text{subject to} & 0 \leq p_j \leq p_{\text{max}}, \quad j = 1, \dots, m \end{array}$$

how to solve?

1. use uniform power: $p_j = p$, vary p
2. use least-squares:

$$\text{minimize } \sum_{k=1}^n (I_k - I_{\text{des}})^2$$

round p_j if $p_j > p_{\text{max}}$ or $p_j < 0$

3. use weighted least-squares:

$$\text{minimize } \sum_{k=1}^n (I_k - I_{\text{des}})^2 + \sum_{j=1}^m w_j (p_j - p_{\text{max}}/2)^2$$

iteratively adjust weights w_j until $0 \leq p_j \leq p_{\text{max}}$

4. use linear programming:

$$\begin{array}{ll} \text{minimize} & \max_{k=1, \dots, n} |I_k - I_{\text{des}}| \\ \text{subject to} & 0 \leq p_j \leq p_{\text{max}}, \quad j = 1, \dots, m \end{array}$$

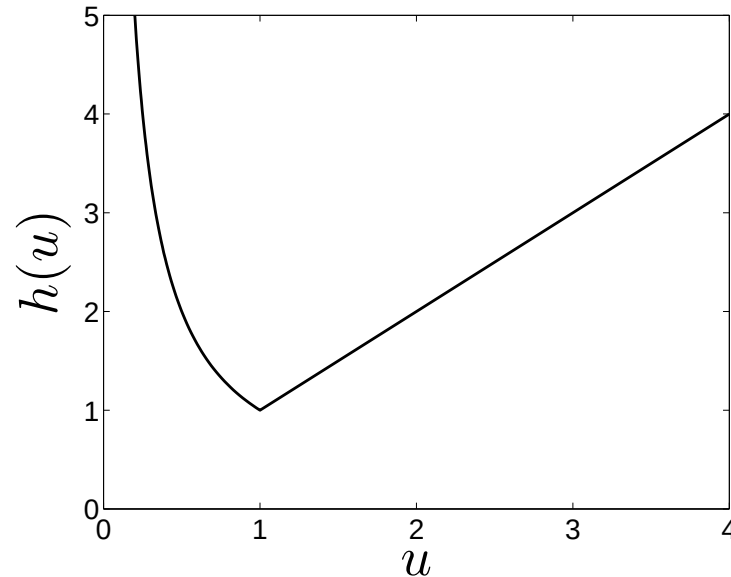
which can be solved via linear programming

of course these are approximate (suboptimal) ‘solutions’

5. use convex optimization: problem is equivalent to

$$\begin{array}{ll} \text{minimize} & f_0(p) = \max_{k=1,\dots,n} h(I_k/I_{\text{des}}) \\ \text{subject to} & 0 \leq p_j \leq p_{\max}, \quad j = 1, \dots, m \end{array}$$

with $h(u) = \max\{u, 1/u\}$



f_0 is convex because maximum of convex functions is convex

exact solution obtained with effort $\approx$ modest factor $\times$ least-squares effort

additional constraints: does adding 1 or 2 below complicate the problem?

1. no more than half of total power is in any 10 lamps

2. no more than half of the lamps are on ($p_j > 0$)

- answer: with (1), still easy to solve; with (2), extremely difficult
- moral: (untrained) intuition doesn't always work; without the proper background very easy problems can appear quite similar to very difficult problems

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- Need to know: represent the problem, relation between solution and dual, optimality condition, some analysis skills etc..

Using Optimization Tools in Your Research

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- More difficult problems (nonlinear or combinatorial optimization):
 - Locally optimal solutions in polynomial time;
 - Heuristic algorithms (simulated annealing, genetic algorithms, etc.): no guaranteed optimality in polynomial time.

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In ECEN 629

- Theory: useful for the first three kinds of situations;
- Applications: emphasis on electrical engineering (comm., SP, learning)
- Algorithms: understand how some algorithms work (not the precise analysis)

A Little Mathematical Background and Notation

Algebraic form:

- A vector $x \in \mathbf{R}^n$: $x = (x_1, x_2, x_3, \dots, x_n)$;
- Multiplying a vector with a scalar: $\alpha x = (\alpha x_1, \alpha x_2, \alpha x_3, \dots, \alpha x_n)$;
- Addition of two vectors: $x + y = (x_1 + y_1, x_2 + y_2, x_3 + y_3, \dots, x_n + y_n)$;
- Additive inverse: $-x = (-x_1, -x_2, -x_3, \dots, -x_n)$;
- Inner product: $\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i$;
- Euclidean norm (ℓ_2 norm): $\|x\|_2 = (x^T x)^{1/2} = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}$;
- Cauchy-Schwartz inequality: $|x^T y| \leq \|x\|_2 \|y\|_2$.

Geometric interpretation: usually in 2 or 3 dimension

- Addition of two vectors;
- Multiplying with a scalar;
- Multiplication of two vectors and Cauchy-Schwartz inequality.

Example question: $\max_{x \in \mathbf{R}^n; \|x\| \leq 1} a^T x$?

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- For optimization with continuous functions, $<$ may be the same as $\leq$;
- Solution may not exist, unless the optimality occurs in the interior.

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 - Pareto optimal: cannot decrease one without increasing another;
- You can fix some first and optimize the remaining one;
- Or choose an objective as the weighted summation of them.

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- Both the objective function and the constraint functions are linear

$$f_i(\alpha x + \beta y) = \alpha f_i(x) + \beta f_i(y), \text{ for all } x, y \in \mathbf{R}^n, \alpha, \beta \in \mathbf{R}. \quad (1)$$

- Otherwise called non-linear.

Role of Convex Optimization in Nonconvex Problems

Relax the original problem into a convex problem for:

- Initialization for local optimization;
- The convexity requirements can be relaxed in some limited cases;
- Convex heuristic for nonconvex optimization: Some algorithm designed for convex optimization works surprisingly well for nonconvex problem;
- Bounds for global optimization;

Brief history of convex optimization

theory (convex analysis): ca1900–1970

algorithms

- 1947: simplex algorithm for linear programming (Dantzig)
- 1960s: early interior-point methods (Fiacco & McCormick, Dikin, . . .)
- 1970s: ellipsoid method and other subgradient methods
- 1980s: polynomial-time interior-point methods for linear programming (Karmarkar 1984)
- late 1980s–now: polynomial-time interior-point methods for nonlinear convex optimization (Nesterov & Nemirovski 1994)

applications

- before 1990: mostly in operations research; few in engineering
- since 1990: many new applications in engineering (control, signal processing, communications, circuit design, . . .); new problem classes (semidefinite and second-order cone programming, robust optimization)